

- Instruction:
- 1) The Progmocds tab can be copied (as required) to create multiple Progmocds tabs, according to the programming module in the study
 - 2) Update the "Title" of the Progmocds to be descriptive (e.g. 5A-Baseline Characteristics)
 - 3) List the newly added Progmocds (table) in Section C of the Cover tab
 - 4) Delete any examples in grey /italic text that are not relevant/applicable to the study once the programming specification has been finalized

List the newly added ProgMods table(s) in Section C of the Cover tab																
(4) Delete any examples in grey/italic text that are not relevant/applicable to the study once the programming specification has been finalized																
Title10.LOT1_BASE_END																
Section A: Defining Medication Available Period (MAP) medications																
Time Period	Variable	Label	Values	Definition	Code Lists Group	Additional Notes	Date Modified (Initials + Date)	QC Reviewed (Initials + Date)								
	LOT1_END_DT_TEMP	Temporary end date for LOT1 base period		The LOT will continue until the earliest of any of the following: (1) Discontinuation of all agents in the regimen (with or without switch to a new agent) - the LOT end date is based on the run-out date, defined as the last date that any MM medication in that LOT is considered to be available; (2) Unplanned SCTs - if an autologous SCT occurs > 180 days following a previous autologous SCT, the LOT will end the day before the later SCT; any allogeneic SCTs are considered a new LOT; (3) SCTs not followed by maintenance within 180 days - the last day of the LOT is the date of the SCT; (4) Death; (5) Health plan disenrollment; (6) End of the study period; (7) End of maintenance regimen. Note: Substitution of a biologic reference product with any of its biosimilars, or between biosimilars of the same reference product, does not advance the LOT (Rule 1 - this is an exception, not an ending event).		LOT1_BASE_END_DT derived from LOT1_BASE_DISCON_DT (max MAP_END_DT from T_MEDICAL_PROC_CD/INDC and T_RX_NDC for induction meds), LOT1_BASE_IST_ADD_MED_DT (first MAP_START_DT from T_MEDICAL_U_RX for non-induction meds minus 1), LOT1_TX_ENDDATE (earliest LOT-ending SCT from T_MEDICAL procedure codes minus 1), and OBS_END_DT (LEAST of study_end from config, DEATH_DT from T_DOD YMODD, disenrollment from T_MEMBER_COUNT_ENROLLMENT_ELIGEND).		JM 15Apr/2026								
FU_PD																
	LOT1_END_REASON_TEMP	Temporary LOT1 base period end reason		The LOT ends for one of these reasons: (Rule 2) Discontinuation of all agents in the regimen, with or without switch to a new agent; (Rule 2) If the current LOT is immediately interrupted by a new agent, the LOT end date is the day before the first administration/dispense date of the new agent; (Rule 3) Unplanned SCTs - if an autologous SCT occurs > 180 days following a previous autologous SCT, or any allogeneic SCT; (Rule 4) SCTs not followed by maintenance within 180 days; (Rule 5) Death; (Rule 6) Health plan disenrollment; (Rule 7) End of the study period; (Rule 8) End of maintenance regimen.		End reason derived from matching the LOT1 end date against dates from: T_MEDICAL_U_RX (discontinuation via MAP_END_DT), T_MEDICAL_U_RX (new med via MAP_START_DT), T_DOD (death via YMODD), T_MEDICAL (SCT via procedure codes in T_MED_PROCEDURE), ENDDATE from T_MEMBER_COUNT_ENROLLMENT (disenrollment) and study_end config.		JM 06Apr/2026								
FU_PD																
	LOT1_HSCOT_AUTO_SING_FLAG	Flag: base period had single valid autologous HSCOT		Stem cell transplant (SCT): A LOT-specific binary variable is created to classify whether patients have an SCT during the LOT. SCT is operationalized as > 1 procedure code for an SCT. A single autologous SCT is one that is not part of a valid tandem pair. Single autologous SCTs and tandem SCTs are not unplanned SCTs and are considered a continuation of the line of therapy.		Autologous HSCOT identified from T_MEDICAL procedure codes (PROC_CD, BILL_PROC_CD) matching SCT codelist entries for AUTO type. Claims grouped by 14-day windows from T_MEDICAL_FST_DT. Gap validation (60 days) and tandem detection (180 days) applied to grouped dates.		JM 06Apr/2026								
FU_PD																
	LOT1_HSCOT_AUTO_TAND_FLAG	Flag: base period had tandem valid autologous HSCOT		Patients will be classified as having a planned autologous tandem SCT if they receive a second autologous SCT more than or equal 60 days and less than or equal 180 days after the first planned autologous SCT within the same line of therapy (LOT). If the interval between the two autologous SCTs falls outside this window, the sequence will be considered unplanned. An autologous SCT followed by an allogeneic SCT will not be considered planned, because allogeneic SCTs typically follow failed autologous SCTs.		Tandem detection uses T_MEDICAL procedure codes for both AUTO events. The 180-day window is applied to grouped AUTO dates (after 14-day windowing and 60-day gap validation). ALLO events between AUTO_DT_1 and AUTO_DT_2 checked via T_MEDICAL.		JM 15Apr/2026								
FU_PD																
	LOT1_TX_AUTO_MAX_DT	Latest date of patient's valid HSCOT		This variable captures the date of the second valid tandem autologous (AUTO) SCT -- where a tandem is two AUTOs >= 60 and <= 180 days apart -- or, if only a single AUTO occurred, the date of that single AUTO.		Derived from T_MEDICAL procedure code dates for autologous HSCOT, after 14-day windowing and 60-day gap validation.		JM 15Apr/2026								
FU_PD																
	LOT1_TX_AUTO_FLAG	Flag: patient had any valid autologous HSCOT		Stem cell transplant (SCT): A LOT-specific binary variable is created to classify whether patients have an SCT during the LOT. This binary variable indicates whether the patient received any valid autologous transplant, regardless of single or tandem.		Derived flag from T_MEDICAL procedure code analysis for autologous HSCOT events.		JM 06Apr/2026								
FU_PD																
	LOT1_BASE_MEEDS	LOT1 induction medications		The LOT1 regimen is defined as all multiple myeloma (MM) oncology therapies received on and within 60 days of the LOT1 start date, with the first therapy received considered the induction therapy. This variable therefore captures the full set of oncology agents administered during this 60-day window, excluding non-oncology agents such as corticosteroids. Switches between a biologic reference product and its biosimilar(s), or between biosimilars of the same reference product, are permitted and do not constitute a change in regimen or advance the line of therapy (Phaduri, 2021).		Medications identified from T_MEDICAL (PROC_CD, BILL_PROC_CD, NDC) and T_RX (NDC) matched against CL_MMA_CODELIST (130-41). Medication abbreviations from CL_MMA_ROLLUP (130-40). Claims filtered by MAP_START_DT within induction window from LOT1_START_DT.		JM 06Apr/2026								
FU_PD																
	LOT1_BASE_END_DT	Final LOT1 base period end date		The LOT will continue until the earliest of any of the following (Rules 2-8 are ending events; Rule 1 from the protocol defines permissible substitutions that do NOT end the LOT): "Per Rule 2: "The LOT end date is based on the run-out date, defined as the last date that any MM medication in that LOT is considered to be available." If the current LOT is immediately interrupted by a new agent, the LOT end date is the day before the first administration/dispense date of the new agent."		LOT1_BASE_END_DT derived from priority evaluation of dates from T_MEDICAL_U_RX (MAP_END_DT for discontinuation), MAP_START_DT for new med), T_MEDICAL procedure codes (SCT events), T_DOD (YMODD for death), T_MEMBER_COUNT_ENROLLMENT (ELIGEND for disenrollment), and study_end config parameter.		JM 06Apr/2026								
FU_PD																
	LOT1_BASE_END_REASON	Final LOT1 base period end reason		LOT1 ends for one of the following reasons (Rules 2-8 from the protocol): (2) Discontinuation of all agents; (3) Unplanned SCTs; (4) SCTs not followed by maintenance within 180 days; (5) Death; (6) Health plan disenrollment; (7) End of study period; (8) End of maintenance regimen. Rule 1 (permissible substitutions - biologic reference product with biosimilars) does not advance the LOT and is not an ending event.		End reason derived from evaluation of dates across T_MEDICAL_U_RX (indication events), T_MEDICAL (SCT procedure codes), T_DOD (death), and T_MEMBER_COUNT_ENROLLMENT (disenrollment). Priority logic applied analytically.		JM 06Apr/2026								
FU_PD																
	FIRST_ALLO_DT	Date of first allogeneic HSCOT during LOT1		Allogeneic SCT: Presence of an allogeneic SCT immediately ends a LOT and starts a new LOT. Rule 3: Any allogeneic SCTs are considered a new LOT and the current LOT will end the day before an allogeneic SCT occurs. "Allogeneic SCT: Presence of an allogeneic SCT is classified as its own LOT, with no other MM		ALLO SCT identified from T_MEDICAL procedure codes matching CL_SCT_CODELIST entries where SCT_TYPE="ALLO". Date from FST_DT in T_MEDICAL. Sequential ordering applied to identify first ALLO event.		JM 06Apr/2026								
FU_PD																
<	>	1.Cover	2.QC Review	3.Data Prep	4.Study/Pop	5A.MMA_MED	5B.MAP_MED	6.LOT1_BASE	7.SCT	10.LOT1_BASE_END_OK	40.CL_MMA_ROLLUP	41.CL_MMA_CODELIST	41a.MMA_NDC_CODELIST	41b.MMA_HCRCS_CODELIST	4	...

A	B	C	D	E	F	G	H							
FU_PD	FIRST_ALLO_DT	Date of first allogeneic HSCT during LOT1		Allogeneic SCT: Presence of an allogeneic SCT immediately ends a LOT and starts a new LOT. Rule 3: Any allogeneic SCTs are considered a new LOT and the current LOT will end the day before an allogeneic SCT occurs. 'Allogeneic SCT: Presence of an allogeneic SCT is classified as its own LOT, with no other MM therapies included in that LOT.'		ALLO SCT identified from T_MEDICAL procedure codes matching CL_SCT_CODELIST entries where SCT_TYPE=ALLO. Date from FST_DT in T_MEDICAL. Sequential ordering applied to identify first ALLO event.	JM105Apr/2026							
FU_PD	FIRST_CART_DT	Date of first CAR-T therapy during LOT1		CAR-T: CAR-T cellular therapy infusions are classified as their own LOT. Therapies (including supportive agents, e.g., corticosteroids) given within 45 days of CAR-T cellular therapy are consolidated as part of the CAR-T LOT.		CAR-T identified from T_MEDICAL procedure codes matching CL_SCT_CODELIST entries where SCT_TYPE=CART. Date from FST_DT in T_MEDICAL.	JM115Apr/2026							
FU_PD	LOT_DISCON_GAP_DAYS	Discontinuation gap threshold (days)		Discontinuation of the LOT1 regimen occurs when all MM oncology agents in the line of therapy are stopped, if the regimen is discontinued and no new agent is initiated, the LOT end date is defined as the run-out date of the last administered agent. Sensitivity analyses will assess the robustness of the line of therapy rules by varying key algorithm parameters.		Gap calculated as datediff(OBS_END_DT, RAW_DISCON_DT) where OBS_END_DT is derived from T_MEMBER_CONT_ENROLLMENT (ELCEND), T_DOD (YMOBOD), and study_end config RAW_DISCON_DT is max(MAP_END_DT) from T_MEDICAL_RX for induction medications.	JM115Apr/2026							
FU_PD	MEDICAL_DAY_SUPPLY	Assumed days supply for medical claims		The day's supply or medical claims is always missing and therefore a 28-day supply is assumed. When pharmacy claims are missing days supply, or when days supply values are anomalous, a 28-day supply is assumed.		Applied to T_MEDICAL claims where DAYS_SUPPLY is typically populated but when missing/anomalous, 28 days is substituted. The assumed supply extends MAP_END_DT = FST_DT (or FILL_DT) + days_supply - 1.	JM106Apr/2026							
FU_PD	LOT1_BASE_LENGTH	LOT1 base period duration in days		The protocol does not define LOT1_BASE_LENGTH as a named variable. Duration is implied by Protocol Section 5.1.1, which defines LOT1 start date (LOT1 begins on the date of the first fill for an MM therapy following a patient's index date) and LOT end date (earliest of any of its biosimilars, or between biosimilars of the same reference product, does not advance the LOT; such permissible substitutions are considered within the same line, which is named according to the drug given for the longest duration).		Duration computed from dates already derived from T_MEDICAL/T_RX (LOT1_START_DT from first MAP_START_DT) and LOT1_BASE_END_DT (priority-based end date).	JM106Apr/2026							
FU_PD	PERMISSIBLE_SUBS_EFFECT	Permissible substitution rules affecting LOT1 end		Substitution of a biologic reference product with any of its biosimilars, or between biosimilars of the same reference product, does not advance the LOT; such permissible substitutions are considered within the same line, which is named according to the drug given for the longest duration.		Substitutions tracked via CL_MMA_ROLLUP (Tab 4b) which maps medication abbreviations. The permissible_subst referenceable pairs original_med with substitute_med. When a new medication claim appears in T_MEDICAL/T_RX, it is checked against permissible_subst before being classified as a new medication addition.	JM106Apr/2026							
FU_PD	LOT1_TX_ENDDATE_REASON	SCT end reason type (AUTO/ALLO/CART)		The type of stem cell transplant (SCT) determines how LOT1 ends. In the case of an autologous SCT, an unplanned transplant ends the current line of therapy the day before the SCT date. For an allogeneic SCT, the current LOT1 ends the day before the allogeneic SCT date, and a new LOT1 begins from the date of the transplant. CAR-T cellular therapy infusions are classified as their own line of therapy, with the preceding LOT1 ending the day before the CAR-T infusion date; however, if a new oncology agent is introduced within 45 days of the CAR-T infusion, that agent is incorporated into the CAR-T LOT rather than initiating a new line of therapy.		SCT type determined from T_MEDICAL procedure codes matched against CL_SCT_CODELIST (Tab 4a) which classifies procedures as AUTO, ALLO, or CART. Priority is earliest LOT-ending date wins.	JM115Apr/2026							
FU_PD	LOT1_TX_AUTO_DT_1/ LOT1_TX_AUTO_DT_2	Dates of first and second valid autologous HSCT		Planned Tandem SCT: Patients will be classified as having a planned Tandem SCT if they receive at least two autologous SCTs >= 60 to <= 180 days apart. 'SCT is operationalized as >1 procedure code for an SCT.' Tandem SCTs are two SCTs >= 60 to <= 180 days apart. These are considered planned and a continuation of the line of therapy. Codes for SCTs are in the programming specification document.		AUTO HSCT dates derived from T_MEDICAL procedure codes (PROC_CD, BILL_PROC_CD) matched against CL_SCT_CODELIST where SCT_TYPE=AUTO. FST_DT from T_MEDICAL is the raw claim date, 14-day windowing groups claims; 60-day gap validation separates distinct events.	JM106Apr/2026							
FU_PD	LOT1_BASE_DISCON_DT	LOT1 base period discontinuation date		Discontinuation of the regimen occurs when all MM agents in the LOT are stopped, either without initiation of a new MM agent (all agents have reached their run-out dates and no further MM administration/dispenses are observed), or with initiation of a new MM agent that was not present in the induction regimen. The LOT end date is based on the run-out date, defined as the last date that any MM medication in that LOT is considered to be available.		max(MAP_END_DT) derived from T_MEDICAL_FST_DT + medical_day_supply - 1 and T_RX (FILL_DT + DAYS_SUP - 1) for induction medications matched via CL_MMA_CODELIST. Gap validated against OBS_END_DT from T_MEMBER_CONT_ENROLLMENT/T_DOD.	JM106Apr/2026							
FU_PD	LOT1_BASE_1ST_ADD_MED T/LOT1_BASE_1ST_ADD_MED	Date and name of first non-induction medication added		If the current LOT1 is immediately interrupted by a new agent (the new agent starts on or before the run-out date of the current regimen), the LOT end date is the day before the first administration/dispense date of the new agent. Per Rule 1: permissible substitutions (biologic reference product with biosimilars) do not advance the LOT.		First non-induction medication identified from T_MEDICAL (PROC_CD, BILL_PROC_CD, NDC) and T_RX (NDC) matched against CL_MMA_CODELIST (Tab 4c) where the medication is NOT in base_meds (induction + permissible subs). MAP_START_DT from MAP algorithm.	JM106Apr/2026							
FU_PD	INDUCTION_WINDOW_DAYS	Induction regimen identification window (days)		The LOT1 regimen includes all MM therapies received within 60 days (Madduri, 2021) on and following the LOT1 start date. For LOT2-LOT5, the LOT1 regimen includes all MM therapies received within 30 days on and following the LOT1 start date.		Applied to MAP_START_DT from T_MEDICAL/T_RX claims. Medications with MAP_START_DT within the window are classified as induction via CL_MMA_CODELIST/CL_MMA_ROLLUP matching.	JM106Apr/2026							
FU_PD	ALLO_ALWAYS_ENDS_LOT	Allogeneic SCT always ends current LOT (protocol rule)		Allogeneic SCT: Presence of an allogeneic SCT immediately ends a LOT and starts a new LOT. An autologous SCT followed by an allogeneic SCT will not be considered planned because an allogeneic SCT typically follows failed autologous SCTs. Rule 3: Any allogeneic SCTs are considered a new LOT and the current LOT will end the day before an allogeneic SCT occurs. 'Allogeneic SCT: Presence of an allogeneic SCT is classified as its own LOT, with no other MM therapies included in that LOT.'		ALLO SCT identified from T_MEDICAL procedure codes matched against CL_SCT_CODELIST where SCT_TYPE=ALLO. Date from FST_DT in T_MEDICAL.	JM106Apr/2026							
FU_PD	CART_45D_CONSOLIDATION	CAR-T 45-day supportive agent consolidation rule		CAR-T cellular therapy infusions are classified as their own line of therapy. Oncology therapies administered within 45 days of the CAR-T infusion are consolidated as part of the CAR-T LOT.		Would require checking T_MEDICAL/T_RX claims within 45 days of FIRST_CART_DT and reclassifying them from LOT1 to the CAR-T LOT. Affects MAP_END_DT for induction meds and MAP_START_DT for potential new MMA-naming convention only.	JM115Apr/2026							
FU_PD	NAMING_DISCREPANCY_SCT_VS_HSCT	Variable naming: spec uses HSCT prefix; code uses SCT prefix		The protocol uses 'SCT' (Stem Cell Transplant) terminology consistently throughout. Stem cell transplant (SCT): The original program spec uses 'HSCT' (Hematopoietic Stem Cell Transplant) prefix for variable names. Both abbreviations refer to the same medical concept.			JM106Apr/2026							
<	>	1.Cover	2.QC Review	3.Data Prep	4.StudyPop	5A.MMA_MED	5B.MAP_MED	6.LOT1_BASE	7.SCT	10.LOT1_BASE_END OK	40.CL MMA ROLLUP	41.CL MMA CODELIST	41A.MMA NDC CODELIST	41B.MMA HCCS CODELIST
If LOT1 contains any of the following in combination with at least one other agent then flag =yes:														

[illegible]